

Department of Artificial Intelligence and Data Sciences



Self Care Website

Submitted By:

Group - 9

B.Tech. CSE-AI 1

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INTRODUCTION

Self Care Website - Guiding users towards mental wellness .

Features :

- 1. Virtual Counselor Chatbot**
- 2. Habit Tracker**
- 3. Books Recommendation**

RESEARCH OBJECTIVES

Primary Goal:

- **To foster mental wellness by creating an all-in-one platform that promotes emotional resilience, habit-building, and continuous learning.**

Novelty:

- **Holistic Approach:** Combines mental health support (Virtual Counsellor), personal habits (Habit Tracker), and continuous learning (Self-Help Book Recommendations) within a single platform.
- **Virtual Counsellor Chatbot:** Offers immediate, anonymous support that encourages users to explore and address their mental well-being in a guided, non-judgmental environment.
- **Preventative Care Emphasis:** Rather than only focusing on treatment, the platform actively encourages proactive self-care through consistent habits and educational resources.

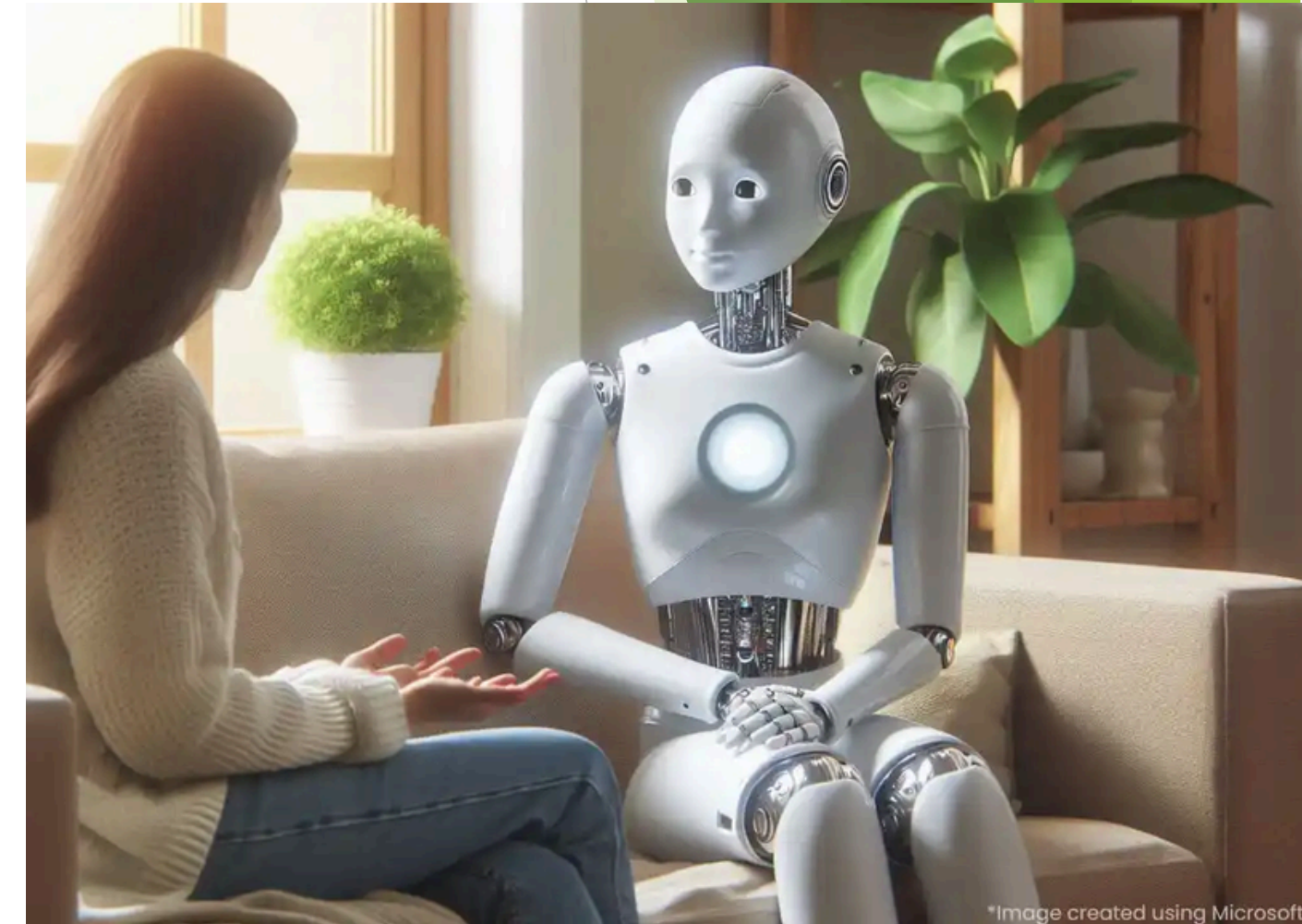
RESEARCH GAPS

Existing Solutions:

- **Limitations:** Many mental wellness apps focus on either therapeutic support (like counseling apps) or habit tracking without integrated mental health guidance.

Innovations in Our Project:

- **Integrated Support:** Combines therapy-like support, habit-building, and learning into one cohesive platform, addressing the need for a unified self-care ecosystem.
- **Trained Chatbot on Real Counselor-Patient Interactions:** Our Virtual Counsellor chatbot is trained on real counselor and patient interactions, which allows it to provide responses grounded in real-life counseling dynamics, making the experience more relatable, empathetic, and effective.



*Image created using Microsoft

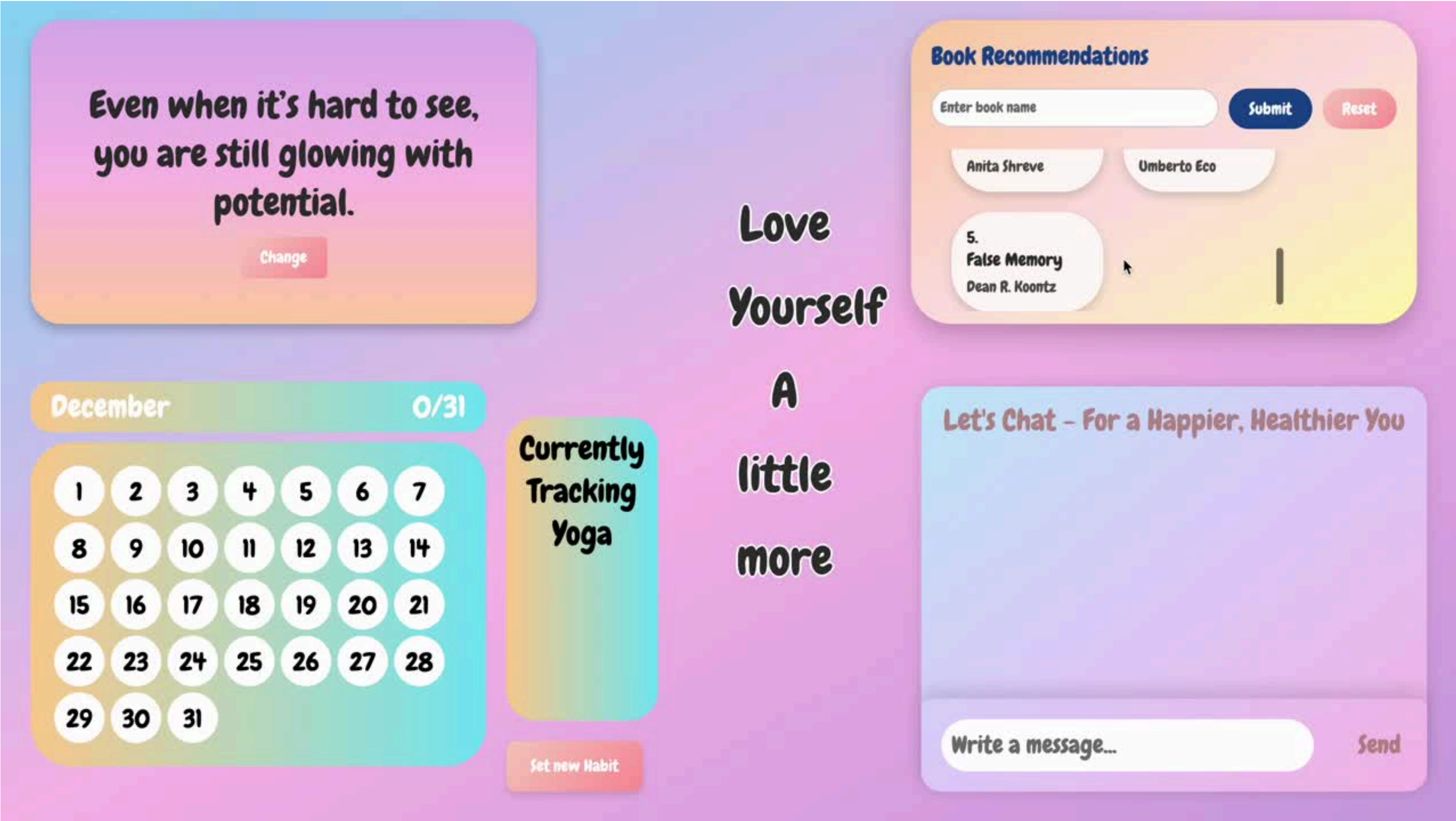
LITERATURE REVIEW

<u>S. No</u>	<u>Author</u>	<u>Title</u>	<u>Published in</u>	<u>Year</u>	<u>Proposed Methodology and Result</u>	<u>Research Gaps</u>
1	Fogg, B.	The Role of Habit Tracking in Promoting Behavioral Change	Behavior Change Review	2009	Goal-setting and habit tracking encourage behavior change and improve adherence to personal improvement routines.	Limited long-term adherence data; potential to integrate habit tracking with personalized guidance for sustained impact.
2	Sucala et al.	Efficacy of Online Counseling for Reducing Anxiety and Depression	Journal of Mental Health (Sample)	2012	Online counseling using CBT techniques is effective in reducing symptoms of anxiety and depression.	Limited personalization; need for empathetic, real-time support to increase user satisfaction and long-term engagement.
3	Gorla et al.	Leveraging Recommendation Algorithms in Mental Health Apps	ACM Transactions on Interactive Intelligent Systems	2013	Using algorithms to recommend resources increases user engagement and promotes self-care in mental health applications.	Need for more granular recommendation algorithms to provide better-suited resources based on real-time mental health assessments.

LITERATURE REVIEW

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4	Ricci et al.	Personalized Book Recommendation Systems for Self-Improvement	Digital Health Applications Journal	2015	Recommendation systems tailored to user needs improve engagement and support personal growth.	Need for more targeted recommendations based on mental health data for better alignment with user goals.
5	Fitzpatrick et al.	Empathy in Chat-Based Mental Health Support	Journal of Medical Internet Research	2017	Chat-based mental health support systems with empathetic responses improve user engagement and overall satisfaction.	Lack of detailed personalization; challenge in creating nuanced responses to complex mental health issues.
6	Clear, J.	Habit Formation and Long-Term Behavioral Change	Atomic Habits	2018	Small, achievable goals with consistent habit tracking lead to sustainable behavior changes and improve self-regulation.	Limited research on habit tracking's impact in digital mental health; further integration with virtual counseling may be beneficial.

Website



Even when it's hard to see,
you are still glowing with
potential.

Change

December

0/31

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Currently
Tracking
Yoga

Set new Habit

Book Recommendations

Enter book name

Submit

Reset

Anita Shreve

Umberto Eco

5.
False Memory
Dean R. Koontz

Love
Yourself

A
little
more

Let's Chat - For a Happier, Healthier You

Write a message...

Send

METHODOLOGY - BUILDING CHATBOT

Dataset : `counsel_chat.csv`

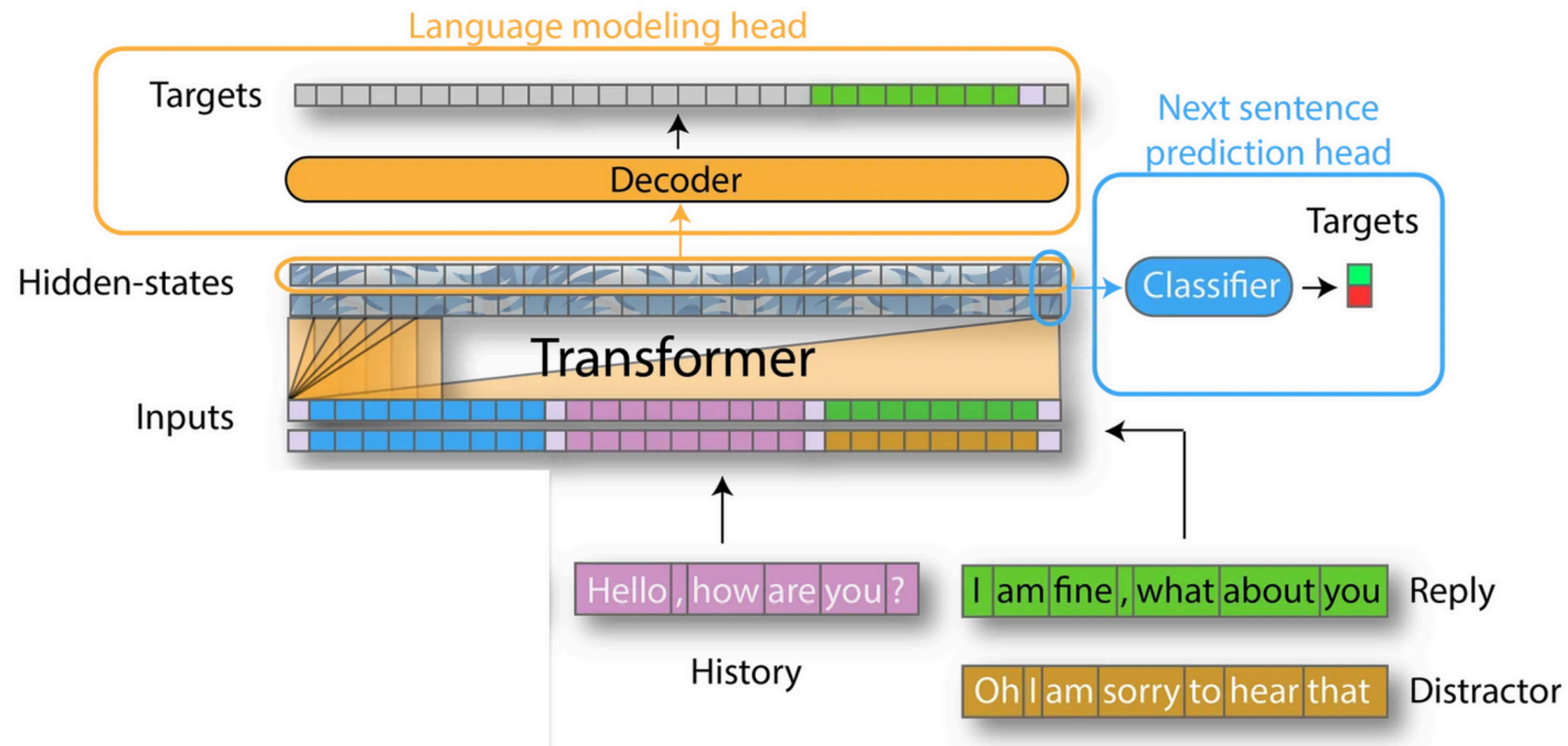
- **Rows - 2129 , columns - 10 .**
- **Columns - `questionID` , `questionTitle` ,`questionText` , `questionLink` , `topic` , `therapistInfo` , `therapistURL` , `answerText` , `upvotes` , `split` .**
- **Number of therapists - 31 .**

20200325_counsel_chat (2)					
	questionID	questionTitle	questionText	answerText	up
0	0	Can I change my feeling of being worthless to everyone?	I'm going through some things with my feelings and myself. I barely sleep and I do nothing I've never tried or contemplated suicide. I've always wanted to fix my issues, but I never How can I change my feeling of being worthless to everyone?	If everyone thinks you're worthless, then maybe you need to find new people to hang out v	
1	0	Can I change my feeling of being worthless to everyone?	I'm going through some things with my feelings and myself. I barely sleep and I do nothing I've never tried or contemplated suicide. I've always wanted to fix my issues, but I never How can I change my feeling of being worthless to everyone?	Hello, and thank you for your question and seeking advice on this. Feelings of worthlesne	

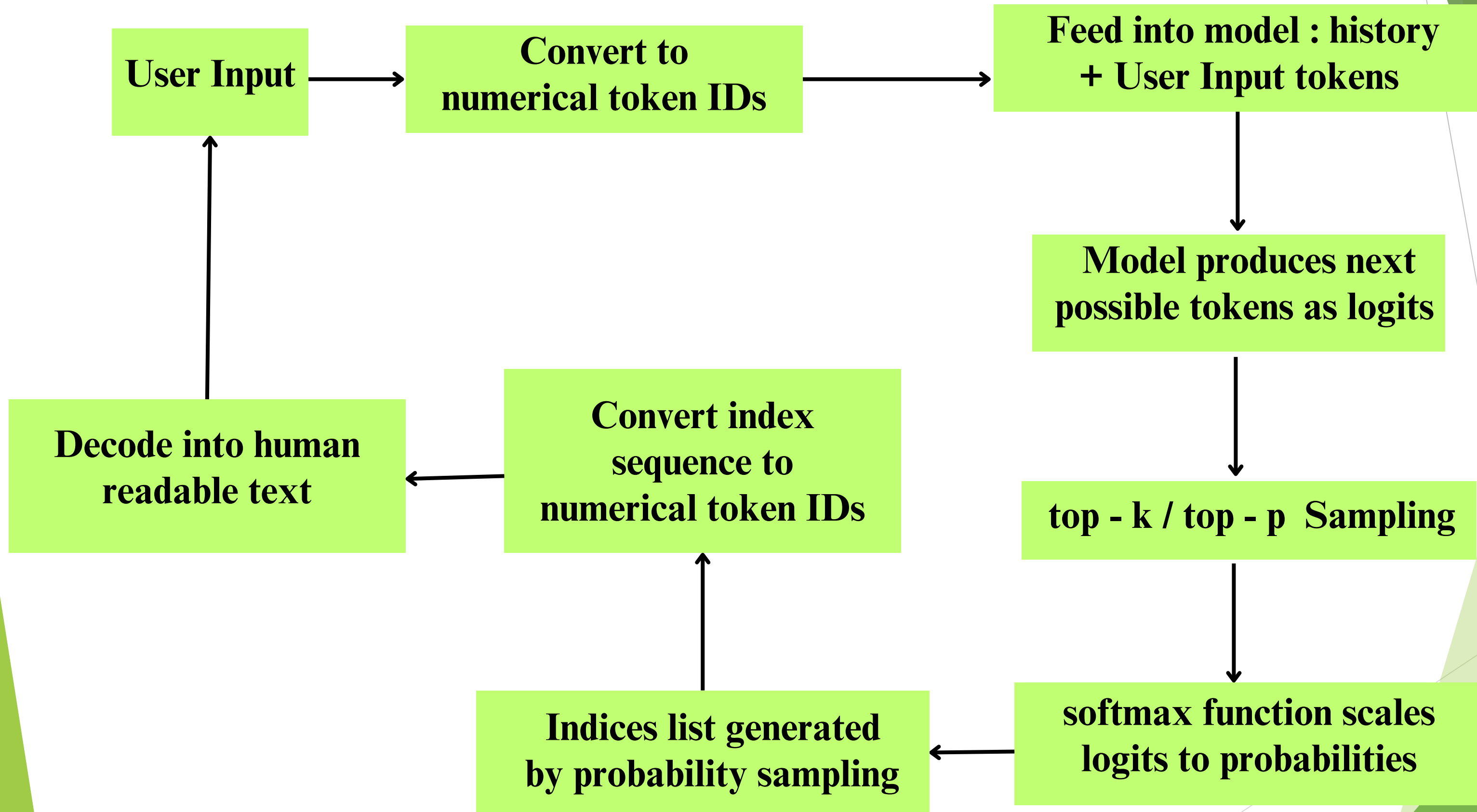
METHODOLOGY

Model Training

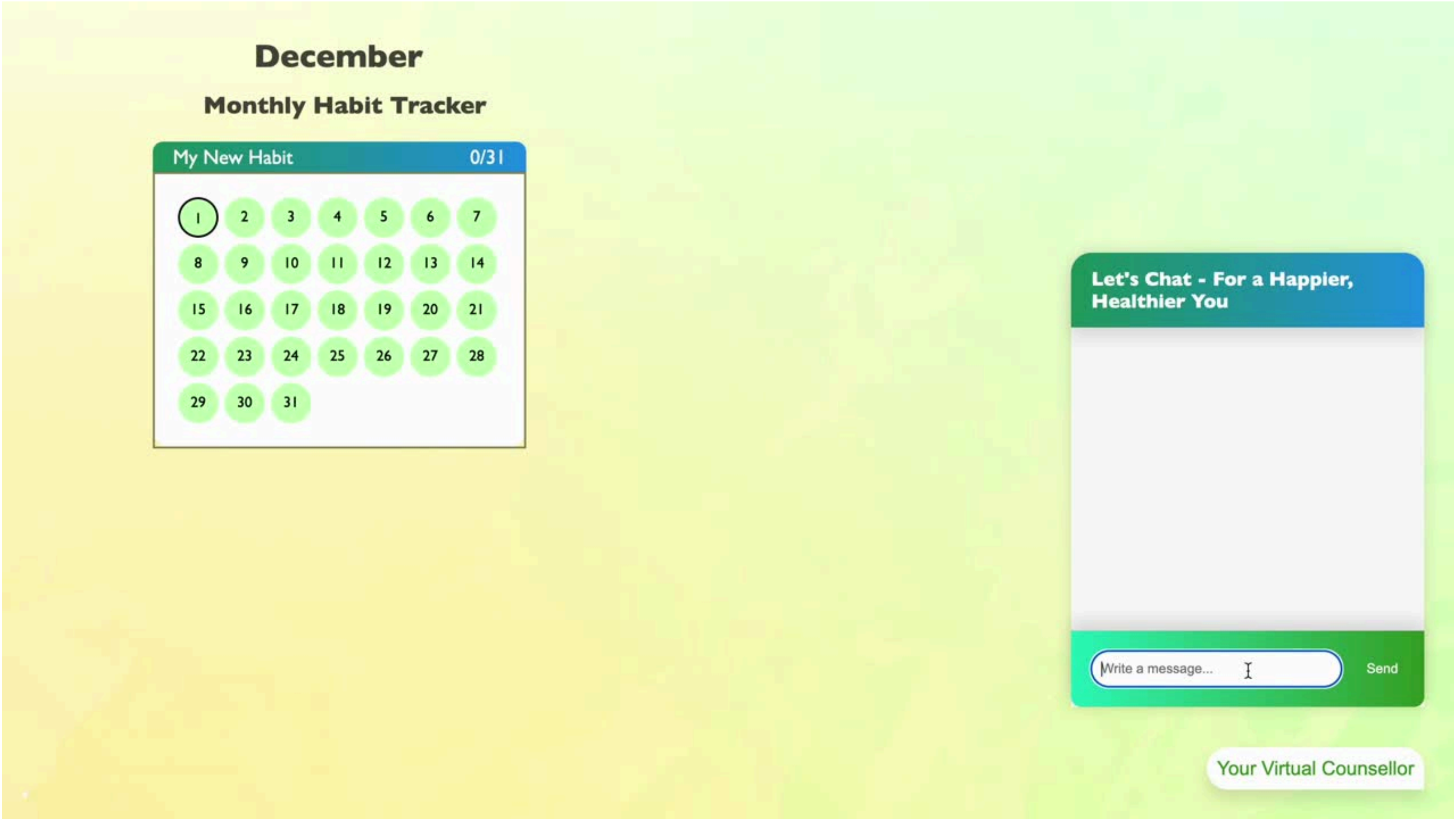
- **Transfer Learning : Used pre-trained model - OpenAI GPT.**
- **Fine tuned the model on counselling dataset.**



CHATBOT FLOW DIAGRAM



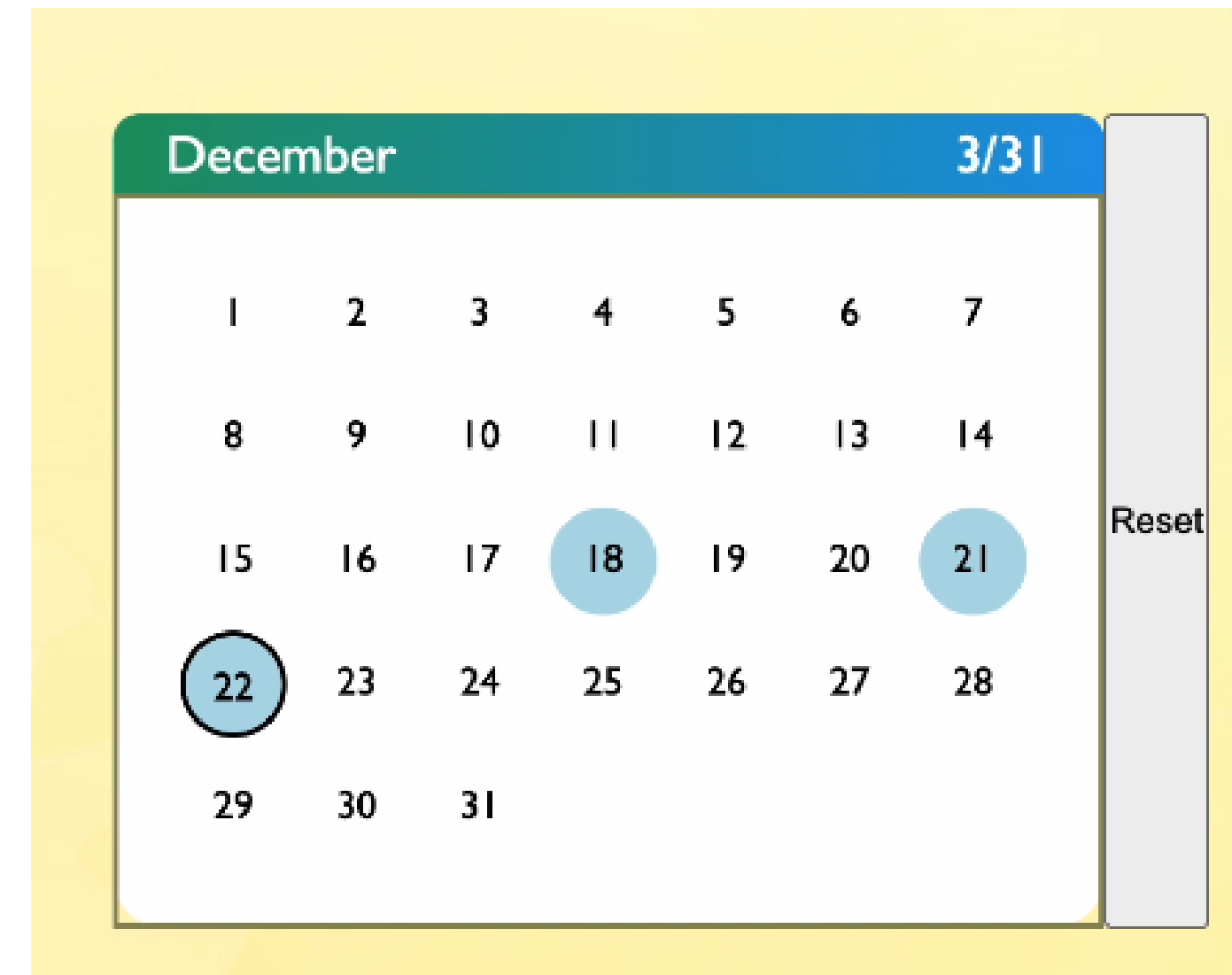
CHATBOT WORKING



Habit Tracker

The habit tracker component works in synergy with the virtual counselor's recommendations:

- **Enables users to set and monitor personal development goals.**
- **Tracks progress over time with visual representations.**
- **Implements suggested lifestyle changes from counseling sessions.**
- **Provides accountability and motivation through progress monitoring.**
- **Features a Flask-based interface for seamless integration with the main platform**



Habit Tracker Working

October

Monthly Habit Tracker

My New Habit0/31

1234567

891011121314

15161718192021

22232425262728

293031

2nd / Check if you're happy with your progress

12345678910111213141516171819202122232425262728293031

12345678910111213141516171819202122232425262728293031

Your Virtual Counsellor

Book Recommendation

This intelligent system enhances the platform's support capabilities by:

- **Analyzing user preferences and mental health goals and find relevant books and journals for personal development.**
- **Tailoring recommendations to specific areas of self-improvement.**
- **Based on the virtual counselor's guidance, user can find reading materials.**
- **Utilizing Flask for seamless integration with the main application**



CONCLUSION

- ***Holistic Wellbeing Platform:*** Our virtual assistant counseling service integrates mental health support, habit tracking, and book recommendations to enhance users' mental, physical, and emotional well-being.
- ***Empathetic Virtual Counseling:*** The Virtual Counselor Chatbot provides personalized and supportive interactions, recommending self-care routines tailored to individual needs.
- ***Structured Goal Tracking:*** The Habit Tracker enables users to set and monitor wellness goals, promoting consistent, positive lifestyle changes.
- ***Curated Book Recommendations:*** (Upcoming) Personalized recommendations for books and journals on mental health and personal development will help users pursue targeted growth.



FUTURE SCOPE

- ***Enhanced Personalization:*** Leveraging AI to refine chatbot interactions and book recommendations based on individual user behavior and preferences.
- ***Expanded Wellness Resources:*** Introduce features such as guided meditations, video tutorials, and live workshops led by mental health experts.
- ***Integration with Wearable Tech:*** Connect the platform with wearable devices to gather health metrics, offering users deeper insights and more customized wellness support.
- ***Multi-Language Support:*** Broaden accessibility by offering counseling and self-care content in multiple languages, catering to a diverse user base.

REFERENCES

- Fogg, B. (2009). *The Role of Habit Tracking in Promoting Behavioral Change*. Behavior Change Review.
- Sucala, M., et al. (2012). *Efficacy of Online Counseling for Reducing Anxiety and Depression*. Journal of Mental Health (Sample).
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- Fitzpatrick, K. K., et al. (2017). *Empathy in Chat-Based Mental Health Support*. Journal of Medical Internet Research.
- Clear, J. (2018). *Habit Formation and Long-Term Behavioral Change*. Atomic Habits.
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THANK YOU